IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-163

1. Topic (12 words max)

Manō : Holistic Framework for Personalized and Community-Driven Cognitive, Emotional, and Resilient Vitality

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	25-26J-163
Year	2025

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Mental health represents a significant global challenge, with approximately 13% of the world's population affected by mental health and substance use disorders, a burden likely intensified by recent global events [1]. The World Health Organization (WHO) indicates that mental disorders contribute to one in six years lived with disability, with severe conditions reducing life expectancy by 10 to 20 years. The economic impact is substantial, surpassing 2.5trillion annually and projected to 6 trillion by 2030 [2]. This highlights a critical gap in accessible, scalable, and privacy-preserving mental health support systems.

Existing digital mental health solutions often face challenges related to data privacy, lack of personalization, and limited integration of diverse support mechanisms. The sensitive nature of mental health data poses significant privacy concerns, making it difficult to collect and utilize real-world datasets for research and model development [3]. While synthetic data generation using Generative Adversarial Networks (GANs) offers a promising avenue to overcome data scarcity and privacy issues, ensuring the fidelity and utility of such synthetic data while maintaining robust privacy guarantees remains a research challenge [4], [5].

Furthermore, many current platforms struggle with providing truly empathetic and context-aware support. Traditional predictive models for mental health often lack the ability to integrate diverse, multimodal data streams (e.g., wearables, surveys, behavioral data) effectively over time, limiting their accuracy and personalization [6], [7]. While Long Short-Term Memory (LSTM) networks show promise in handling such temporal and multimodal data, advancements are needed to enhance their predictive accuracy across diverse user profiles [8]. Similarly, while AI-powered chatbots are increasingly used for mental health support, ensuring their empathy, context-awareness, and ability to provide critical crisis intervention (such as suicide prevention) remains a complex task [9], [10]. Finally, fostering genuine community engagement and dynamically grouping users based on evolving resilience profiles for peer support is an area that requires more sophisticated, adaptive clustering mechanisms.

This project aims to address these multifaceted challenges by developing a holistic framework that leverages cutting-edge machine learning techniques to deliver personalized, privacy-preserving, and community-driven cognitive and emotional health support.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

Manō is a comprehensive, solution designed to enhance mental wellness and cognitive health. It integrates four interconnected machine learning components to provide personalized and community-driven support for stress management, cognitive performance, emotional well-being, resilience, and mindfulness. The system's core components include:

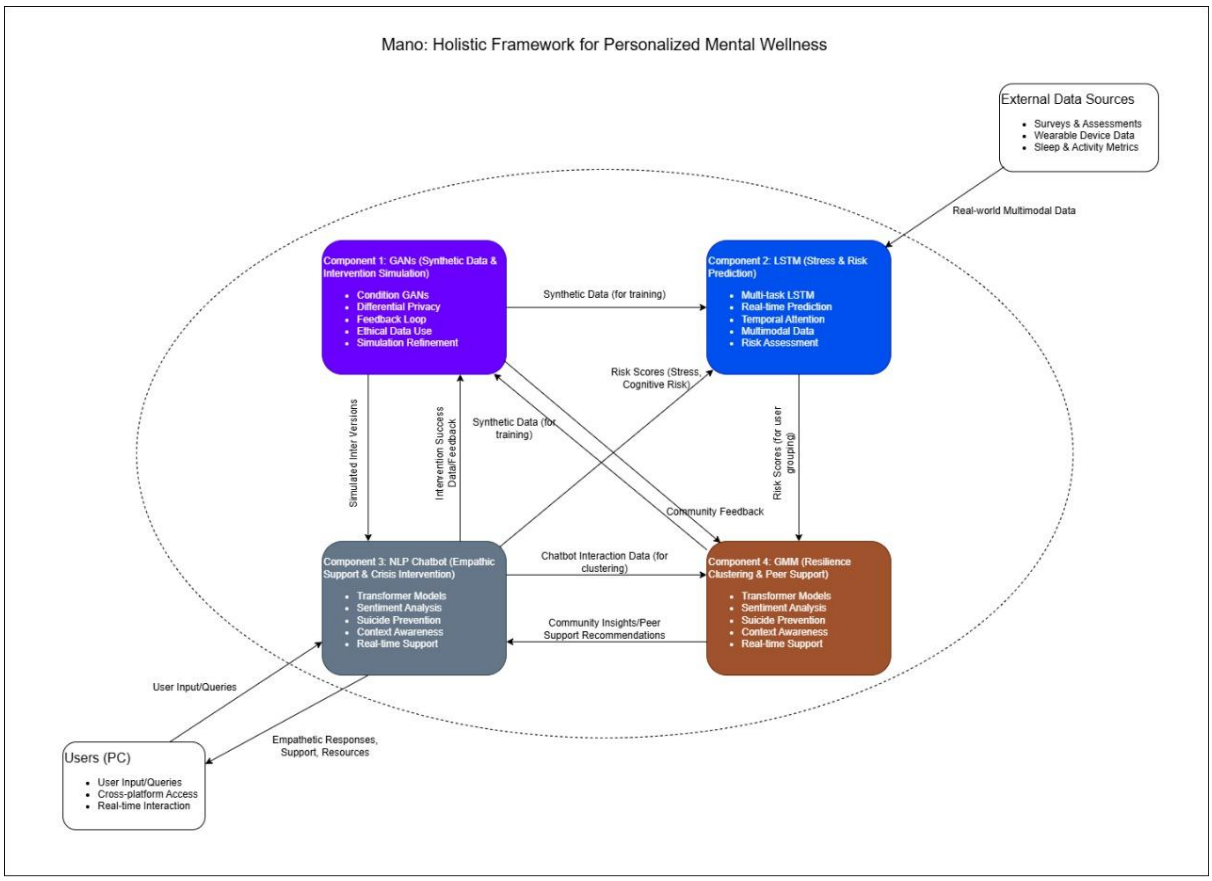
1. **Privacy-Preserving Mental Health Simulation System (GANs):** Generates synthetic mental health data and simulates intervention outcomes, ensuring ethical data use via differential privacy.

2. **Stress and Cognitive Risk Prediction System (LSTM):** Predicts stress levels and cognitive risks using multimodal data, enhanced by a temporal attention mechanism.

3. **Empathetic Conversational Support System (NLP Chatbot):** Offers real-time, context-aware empathetic support, integrating risk scores and simulated interventions, with an added suicide prevention module.

4. **Community-Driven Resilience Clustering System (GMM):** Dynamically clusters users based on resilience indicators to foster peer support.

The conceptual diagram illustrates these four components as distinct yet interconnected modules. Data flows from the GAN (Component 1) to train the LSTM (Component 2), Chatbot (Component 3), and GMM (Component 4). The LSTM feeds risk scores to the Chatbot and GMM. The Chatbot provides interaction data to the GMM and feedback to the GAN. The GMM offers community insights back to the Chatbot and GAN. User interaction points on both PC and mobile devices connect to the central system, demonstrating the cross-platform accessibility and integrated nature of



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

7.The successful execution of the Manō project necessitates a diverse range of specialized domain expertise and knowledge, primarily centered within the Artificial Intelligence and Machine Learning (AI/ML) domain:

Machine Learning: Deep theoretical and practical understanding of advanced machine learning models, including Generative Adversarial Networks (GANs) for data synthesis and simulation, Long Short-Term Memory (LSTM) networks for time-series prediction and multimodal data processing, Natural Language Processing (NLP) with Transformer architectures for conversational AI and sentiment analysis, and Gaussian Mixture Models (GMMs) for unsupervised clustering. Expertise in model training, optimization, and evaluation metrics (e.g., AUC-ROC, statistical similarity).

Data Privacy & Ethics: Critical knowledge of privacy-preserving techniques such as differential privacy, secure data handling practices, and ethical considerations pertaining to sensitive mental health data. This includes understanding bias mitigation in AI systems.

Software Development: Proficiency in full-stack development, specifically React.js and React Native for cross-platform frontend development, Java Spring Boot for building a robust and scalable backend API, and Python for implementing all machine learning components, leveraging its rich ecosystem of libraries (e.g., TensorFlow, Hugging Face Transformers, scikit-learn).

Mental Health Domain: A foundational understanding of mental health concepts, common disorders, stress management techniques, cognitive psychology, resilience theories, and intervention strategies. This knowledge is crucial for designing effective and ethically sound AI-driven support.

Data Requirements: The project relies on a variety of multimodal datasets, including survey responses, wearable sensor data (e.g., heart rate, sleep patterns, activity levels), emotional texts, social media interactions, and behavioral data. Specific datasets identified for training and validation include: Mental Health in Tech Survey (Kaggle), Sleep Health and Lifestyle (Kaggle), Daily Dialog (Hugging Face), Emotion Dataset (Hugging Face), Student Performance (UCI), Human Activity Recognition (UCI), and Social Media and Mental Health (Harvard Dataverse). Crucially, Component 1 will generate synthetic data to augment these real datasets, addressing data scarcity and privacy concerns.

8. Objectives and Novelty

Main Objective The primary objective of the Manō project is to develop a comprehensive, privacy-preserving, and community-driven cross-platform solution that leverages advanced machine learning techniques to enhance cognitive, emotional, and resilient vitality, thereby addressing critical gaps in global mental health care accessibility and personalization.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
WEERASINGHE G.I.D IT22890382	To generate high-fidelity, privacy-preserving synthetic mental health data and simulate the outcomes of various mental health interventions to facilitate research, model training, and intervention prototyping.	* Design and implement a Conditional GAN architecture capable of generating diverse synthetic mental health data (e.g., survey responses, wearable metrics, emotional texts). * Integrate differential privacy mechanisms (e.g., DP-SGD) into the GAN training process to ensure privacy guarantees for synthetic data. * Develop a module to simulate the effects of specific	The novelty lies in the development of a simulation-augmented GAN that not only generates privacy-preserving synthetic mental health data but also models intervention outcomes with a continuous feedback loop from real-world chatbot interactions. This dynamic refinement mechanism, coupled with robust differential privacy, significantly enhances the utility and ethical soundness of synthetic data for mental health research and application development.

		mental health interventions on synthetic user profiles. * Establish a feedback loop mechanism to refine GAN simulations based on real-world interaction data from the chatbot. * Validate the statistical similarity and utility of the generated synthetic data for downstream tasks	
Keerthi KKDD IT22327208	To accurately predict individual stress levels and cognitive performance risks using multimodal data, enabling proactive and personalized interventions.	* Collect and preprocess multimodal data from various sources (surveys, wearables, sleep metrics, synthetic data from Component 1). * Design and implement a multi-task Long Short-Term Memory (LSTM) model for time-series analysis of multimodal data. * Incorporate a temporal attention mechanism within the LSTM to prioritize recent and relevant data points for enhanced prediction accuracy. * Train and validate the LSTM model to predict stress levels and forecast cognitive	The primary novelty is the integration of a temporal attention mechanism within a multi-task LSTM model specifically designed for mental health risk prediction from diverse multimodal data. This mechanism allows the model to dynamically focus on the most indicative recent data, leading to more accurate and personalized risk assessments across varied user profiles, a critical advancement over standard time-series prediction models in this domain.

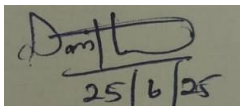
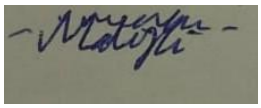
		performance risks (e.g., reduced focus). * Generate interpretable risk scores that can be utilized by other components for tailored interventions.	
Wickramasuriya C.I. IT22890528	To provide real-time, empathetic, and context-aware conversational support for mental well-being, integrating personalized risk assessments and simulated intervention strategies, and offering critical suicide prevention capabilities.	* Develop a transformer-based chatbot architecture, fine-tuned on emotional dialogue datasets. * Implement sentiment analysis capabilities to understand user emotions and conversational nuances. * Integrate real-time risk scores from Component 2 to enable context-aware and personalized responses. * Incorporate simulated intervention strategies from Component 1 into the chatbot's dialogue flow. * Develop and integrate a suicide prevention module with enhanced crisis detection, safety planning prompts, de-escalation techniques, and immediate redirection to emergency services/hotlines. * Ensure ethical guidelines and safety protocols are rigorously applied,	The core novelty lies in the chatbot's dynamic integration of real-time risk scores and simulated interventions to provide highly personalized and evidence-based support, moving beyond generic conversational AI. Furthermore, the explicit development and integration of a comprehensive suicide prevention module with enhanced crisis detection and immediate intervention protocols significantly elevates its utility and ethical impact, making it a pioneering tool for direct mental health crisis support.

		including clear disclaimers and potential human oversight mechanisms.	
Shalinda D.G.M IT22317308	To dynamically identify and group users based on their resilience profiles, fostering supportive peer communities and recommending relevant community activities.	* Identify and extract resilience indicators from diverse data sources (social media, behavioral data, chatbot interactions). * Implement a Gaussian Mixture Model (GMM) for clustering users based on these resilience indicators. * Develop a mechanism for real-time cluster updates to adapt to changes in user profiles and interactions. * Design algorithms to recommend community activities and facilitate peer support within the formed clusters. * Visualize resilience trends within and across communities to promote engagement and provide insights.	The novelty stems from the real-time, adaptive nature of the community formation through continuous GMM updates, directly incorporating dynamic data from chatbot interactions. This allows for the creation of genuinely supportive and relevant peer groups that evolve with user needs, moving beyond static clustering approaches to enable more effective and responsive community-driven mental wellness support.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
WEERASINGHE G.I.D IT22890382	**Component 1: Privacy-Preserving Mental Health Simulation System (GANs)** Specialization Compliance: This component is deeply rooted in cutting-edge AI research, specifically in Generative AI and Privacy-Preserving AI. It demonstrates expertise in designing and implementing complex deep learning architectures (GANs) for synthetic data generation, a core area of modern AI. The integration of differential privacy showcases an understanding of ethical AI and secure machine learning, critical aspects of responsible AI development within a Center of Excellence.
Keerthi K.K.D.D IT22327208	**Component 2: Stress and Cognitive Risk Prediction System (LSTM)** Specialization Compliance: This component exemplifies Predictive AI and Time-Series Analysis within the AI specialization. It involves the application of advanced neural networks (LSTMs) to process multimodal, temporal data for risk assessment. The development of a novel temporal attention mechanism further highlights research and development capabilities in enhancing model performance and interpretability, key areas of focus for an AI Center of Excellence.
Wickramasuriya C.I. IT22890528	**Component 3: Empathetic Conversational Support System (NLP Chatbot)** Specialization Compliance: This component is a direct application of Natural Language Processing (NLP) and Conversational AI, central pillars of the AI specialization. It involves fine-tuning large language models (Transformers) for empathetic dialogue, sentiment analysis, and complex dialogue management. The integration of personalized data and the critical suicide prevention module demonstrates the ability to build sophisticated, ethically responsible, and impactful AI-driven applications, a hallmark of an AI Center of Excellence.
Shalinda D.G.M IT22317308	**Component 4: Community-Driven Resilience Clustering System (GMM)** Specialization Compliance: This component showcases expertise in Unsupervised Learning and Data Mining within the AI specialization. It applies advanced statistical models (GMMs) for dynamic clustering and pattern recognition from diverse data sources. The focus on adaptive community formation and leveraging AI for social impact aligns with the broader goals of an AI Center of Excellence to develop intelligent systems that address real-world societal challenges.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Sanvitha	Kasthuriarachchi	
Co-Supervisor	Ms.	Malithi	Nawarathne	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.

References

[1] S. Graham, "Artificial Intelligence for Mental Health and Mental Illnesses," PMC, 2019. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC7274446/

[2] World Health Organization, "Mental health," 2023. Available: https://www.who.int/news-room/fact-sheets/detail/mental-health

[3] J. M. Górriz, "Computational approaches to Explainable Artificial Intelligence in Neuroscience and Neuroimaging," ScienceDirect, 2023. Available: https://www.sciencedirect.com/science/article/pii/S1566253523002610

[4] "Synthetic Data Generation and Augmentation for Privacy-First Mental Health NLP Models," ResearchGate, May 27, 2025. Available: https://www.researchgate.net/publication/392096763_Synthetic_Data_Generation_and_Augmentation_for_Privacy-First_Mental_Health_NLP_Models

[5] "Generative adversarial networks and synthetic patient data," PMC, 2022. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9345230/

[6] "Synthetic data generation: a privacy-preserving approach to ...," Frontiers in Digital Health, 2025. Available: https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1563991/full

[7] "Integrating Multimodal Data with CNN and LSTM Models," JNS, Mar. 12, 2025. Available: https://www.jneonatalsurg.com/index.php/jns/article/view/2089

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[9] "X-A-BiLSTM: a Deep Learning Approach for Depression Detection in Imbalanced Data," ResearchGate, 2019. Available: https://www.researchgate.net/publication/331416863_X-A-BiLSTM_a_Deep_Learning_Approach_for_Depression_Detection_in_Imbalanced_Data

[10] "The Potential of Chatbots for Emotional Support and Promoting ...," PMC, 2023. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC10625083/

[11] "Chatbots for Mental Health: Leveraging LSTM and Seq2Seq Architectures to Enhance User Well-being," ResearchGate, 2024. Available: https://www.researchgate.net/publication/383330748_Chatbots_for_Mental_health_Leveraging_LSTM_and_Seq2Seq_Architectures_to_Enhance_User_Well-being